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A Neural Machine Translation Approach for Bidirectional Translation between English and Roman Dzongkha

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Abstract

Dzongkha, the national language of Bhutan, is predominantly written using its native script in formal settings. However, in everyday digital communication, many Bhutanese speakers write Dzongkha using Roman characters, an informal practice commonly referred to as Dzonglish. Despite its widespread use in messaging platforms and social media, Dzonglish lacks a standardized orthographic system and has no existing parallel corpus or machine translation system. This paper presents the first study on bidirectional neural machine translation between English and Romanized Dzongkha. We constructed a parallel corpus of 3,813 sentence pairs through a web-based crowdsourced data collection platform. Four neural machine translation architectures trained from scratch were evaluated: Seq2Seq GRU, Seq2Seq with Teacher Forcing, Seq2Seq with Attention, and a Transformer. Additionally, two pretrained MarianMT models (opus-mt-en-mul and opus-mt-mul-en) were fine-tuned for both translation directions. Evaluation was conducted using BLEU, METEOR, chrF, ROUGE-1, ROUGE-2, and ROUGE-L metrics. Results show that from-scratch models struggle on this small low-resource dataset, while MarianMT fine-tuning achieves significantly higher scores, attaining BLEU 5.24 and METEOR 21.29 for English to Dzonglish and BLEU 3.59 and METEOR 17.21 for Dzonglish to English. This work establishes the first baseline NMT system and parallel corpus for English-Dzonglish translation, providing a foundation for future low-resource language technology development in Bhutan.

The dataset, code, and pretrained model checkpoints are publicly available at: <https://github.com/aploslab/Dotu-dzonglish-nmt.git>

Keywords: *Dzonglish, Romanized Dzongkha, Neural Machine Translation, Low-Resource Language, Seq2Seq, Transformer, MarianMT, Bidirectional Translation*

1. Introduction

Bhutan is a small landlocked kingdom in the eastern Himalayas where Dzongkha serves as the national language, used in government institutions, education, and formal communication. While Dzongkha has its own script, the rapid adoption of smartphones and social media has created a practical communication gap. Many Bhutanese speakers, particularly younger generations, find it faster and more convenient to write Dzongkha phonetically using the Roman alphabet on standard keyboards. This informal writing style, commonly called Dzonglish, has become the dominant medium of written communication in digital platforms such as WhatsApp, Facebook, and text messaging.

Despite its pervasive use in everyday digital life, Dzonglish has received virtually no attention from the natural language processing research community. No standardized spelling system exists for Romanized Dzongkha, and the same word may be spelled in multiple ways by different users. This orthographic variability, combined with the absence of a parallel corpus, makes automatic translation between English and Dzonglish a technically challenging and largely unexplored problem.

Existing work on Dzongkha NLP has explored next syllable prediction [1], word segmentation [2], text-to-speech synthesis [3], and most recently, Dzongkha-to-English translation using the native script [4]. However, no prior work has studied translation involving Romanized Dzongkha. The creation of a baseline translation system for Dzonglish is timely given the growing body of literature on low-resource neural machine translation and the practical need for digital language tools that reflect how Bhutanese people actually communicate.

This paper makes three primary contributions. First, we construct and release the first parallel English-Dzonglish corpus, comprising 3,813 sentence pairs collected through a purpose-built web-based crowdsourced platform. Second, we implement and evaluate four neural machine translation architectures trained from scratch: Seq2Seq GRU, Seq2Seq with Teacher Forcing, Seq2Seq with Attention, and a Transformer. Third, we fine-tune two pretrained multilingual MarianMT models for both translation directions and demonstrate that pretrained multilingual representations significantly outperform from-scratch baselines on this low-resource dataset.

The remainder of this paper is organized as follows. Section 2 reviews related literature on neural machine translation, low-resource language processing, and romanization. Section 3 describes the dataset, preprocessing pipeline, and model architectures. Section 4 presents and discusses experimental results. Section 5 concludes with directions for future work.

2. Literature Review

Machine translation has evolved substantially from early rule-based and statistical approaches toward neural methods. Rule-based machine translation (RBMT) requires detailed linguistic knowledge encoded as hand-crafted rules and lexicons, which is difficult to maintain and expensive to scale [5]. Statistical machine translation (SMT) learns translation patterns from bilingual corpora automatically but struggles with sparse data and unseen phrases [6]. Neural machine translation (NMT) overcomes many of these limitations by learning end-to-end representations of source and target languages using deep neural networks.

The sequence-to-sequence framework introduced by Sutskever et al. [7] established the encoder-decoder architecture as the foundation of modern NMT. An encoder network compresses an input sentence into a fixed-length context vector, which a decoder network then uses to generate the translated output word by word. Cho et al. [8] proposed Gated Recurrent Units (GRU) as an alternative to LSTM for capturing sequential dependencies, showing that GRU achieves comparable performance with fewer parameters. The attention mechanism, introduced by Bahdanau et al. [9], substantially improved translation quality for longer sentences by allowing the decoder to attend selectively to different parts of the encoder output at each decoding step, rather than relying on a single compressed context vector.

The transformer architecture, proposed by Vaswani et al. [10], replaced recurrence entirely with multi-head self-attention and positional encoding. By processing all tokens in parallel and modeling global dependencies directly, the transformer achieved state-of-the-art performance on standard translation benchmarks and has since become the dominant architecture in NLP. Teacher forcing [11] is a training strategy for recurrent models in which the ground truth output token from the previous time step is fed as input to the decoder rather than the model's own prediction, leading to faster convergence and more stable training.

Low-resource neural machine translation remains an active research challenge. Xue et al. [12] introduced mT5, a massively multilingual text-to-text transformer pretrained on over 100 languages, demonstrating that multilingual pretraining significantly improves translation quality for languages with limited data. Liu et al. [13] presented mBART, a multilingual sequence-to-sequence denoising autoencoder that can be fine-tuned for translation between language pairs with minimal parallel data. The MarianMT framework, presented by Junczys-Dowmunt et al. [14], provides lightweight transformer-based translation models trained on the OPUS multilingual corpus. The Helsinki-NLP OPUS-MT models [15] built on this framework support translation between English and multiple languages and have been widely used in low-resource translation research due to their efficiency and strong multilingual transfer capabilities.

Code-mixed language processing is directly relevant to Dzonglish, which combines Dzongkha vocabulary expressed in Roman characters, sometimes mixed with English words. Gupta et al. [16] demonstrated that NMT models can learn to translate Hinglish, a code-mixed blend of Hindi and English written in Roman script, showing that neural approaches generalize

to mixed-language inputs. Romanization systems for non-Latin script languages provide another relevant precedent. Knight and Graehl [17] studied neural transliteration between writing systems using sequence-to-sequence models, and standardized systems such as the Revised Romanization of Korean [18] show that informal Roman-script representations of non-Latin languages can be treated systematically. Work by Wangchuk et al. [4] on Dzongkha-to-English translation using the native script represents the closest prior work, achieving a best BLEU score of 64.89 using a transformer on 53,018 sentence pairs. Our work differs in that we address Romanized Dzongkha specifically, operate with a much smaller corpus, and evaluate both translation directions.

3. Method

3.1 System Overview

The experimental pipeline consists of five stages: data collection, preprocessing, vocabulary construction, model training, and evaluation. Figure 1 illustrates the overall system workflow. Two translation directions are studied throughout: English to Dzonglish (EN-DZ) and Dzonglish to English (DZ-EN). All five models are evaluated in both directions using the same dataset split and evaluation metrics.

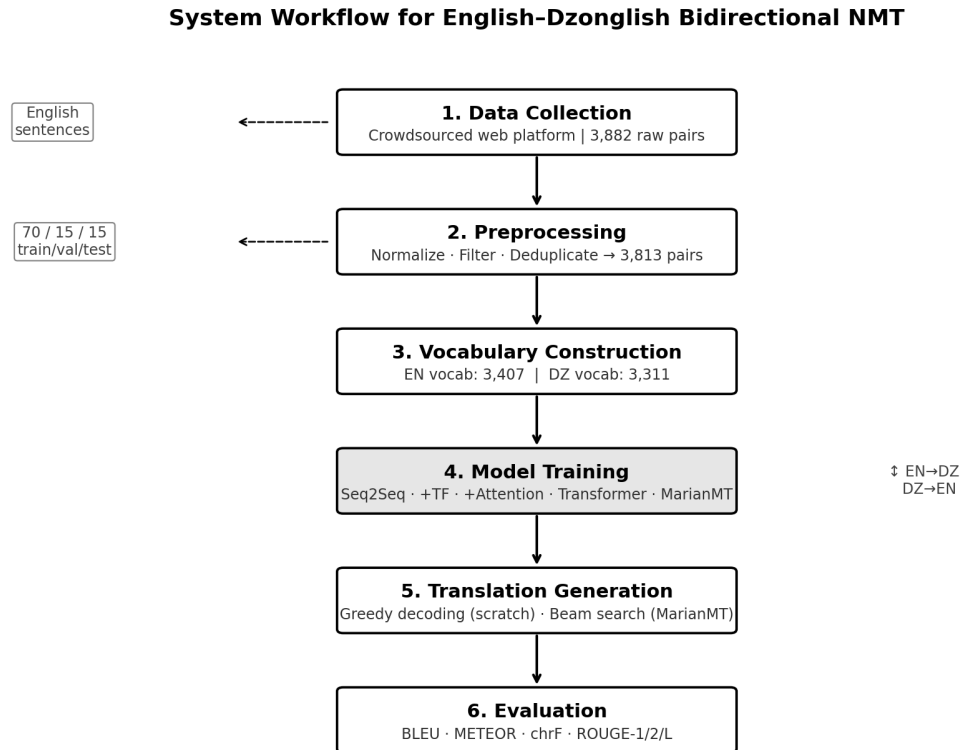


Figure 1. System workflow for bidirectional English-Dzonglish neural machine translation.

3.2 Dataset Collection

No publicly available parallel corpus exists for English and Romanized Dzongkha. To address this gap, we developed a custom web-based data collection platform. The platform presented participants with English sentences and prompted them to type the Dzonglish equivalent in a text field. English source sentences were drawn from common conversational domains including greetings, questions, daily activities, and short declarative statements. The platform was shared with classmates and Bhutanese speakers familiar with Dzonglish, generating translations through crowdsourcing.

The raw collected dataset contained 3,882 sentence pairs. After preprocessing, the final dataset consisted of 3,813 pairs. The corpus covers everyday conversational topics including greetings, questions, directions, weather, food, family, and common daily activities. Example pairs from the dataset are shown in Table 1.

Table 1. Sample English-Dzonglish sentence pairs from the collected corpus.

English Sentence	Dzonglish Translation
watch the rear. i will watch the front.	zhukha ta nga ghi dongkha ta wong
they are doctors.	doctor dhu
he acted like a madman.	kho tsagey zum beydey
it is in the garage.	garage na yoe
i do not have medical training.	nga lu so yi training mi
whose house is this?	ga gi chim mo
tashi will come.	tashi nga wong

3.3 Preprocessing

Preprocessing is critical in low-resource settings because noise in a small corpus has a disproportionate impact on model performance. We applied targeted normalization steps specific to the linguistic properties of each language side, avoiding preprocessing steps that would not benefit NMT.

English normalization involved lowercasing all text to reduce vocabulary size without semantic loss, expanding common contractions (e.g., "don't" to "do not") to improve tokenization consistency, removing URLs and HTML tags that are untranslatable noise, and retaining only alphabetic characters, digits, and basic punctuation. Normalizing whitespace and stripping leading and trailing spaces was applied as a final step.

Dzonglish normalization involved lowercasing, removal of URLs and HTML artifacts, and removal of digit characters. Digits in Dzonglish sentences typically indicate code-mixed or non-conversational content that introduces noise into the translation model. Only alphabetic characters and basic punctuation were retained.

After normalization, we applied four filtering steps. Empty rows were removed. Pairs where the English and Dzonglish sides were identical were discarded as translation errors. Length filtering removed pairs where either side contained fewer than two or more than 40 tokens, eliminating trivially short pairs that provide insufficient training signal and very long pairs that increase memory requirements and training instability. Finally, exact duplicate pairs were removed. Table 2 summarizes the preprocessing pipeline and its effect on the corpus size.

Table 2. Preprocessing pipeline and pair counts at each stage.

Preprocessing Step	Pairs Remaining	Pairs Removed
Raw collected data	3,882	-
After empty removal	3,882	0
After src == tgt removal	3,869	13
After length filter (2-40 tokens)	3,823	46
After deduplication	3,813	10
Final dataset	3,813	69 total

The final dataset of 3,813 sentence pairs was split into training, validation, and test sets using a 70/15/15 ratio, yielding 2,669 training pairs, 572 validation pairs, and 572 test pairs. The split was performed with a fixed random seed for reproducibility, and the three splits were saved to disk to ensure all models were evaluated on identical test sets. Figure 1 shows the sentence length distributions for both languages after preprocessing. English sentences have a mean length of 6.4 words and a maximum of 35, while Dzonglish sentences have a mean of 7.0 words and a maximum of 33.

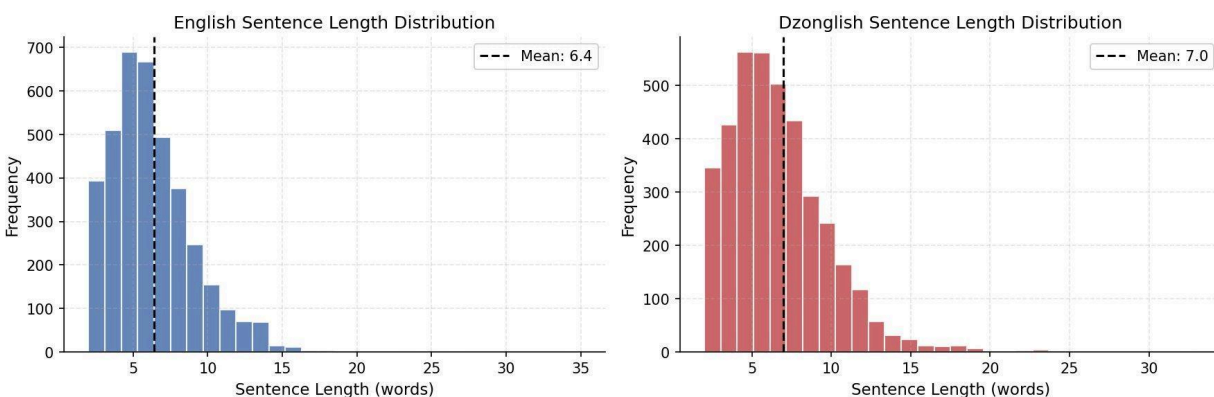


Figure 1. Sentence length distributions for English (left) and Dzonglish (right) after preprocessing. Both languages show a right-skewed distribution with the majority of sentences between 3 and 12 words.

3.4 Vocabulary

Separate vocabularies were built for English and Dzonglish using the training set only, to prevent information leakage from the validation and test sets. Four special tokens were included in each vocabulary: <PAD> for sequence padding, <SOS> for sequence start, <EOS> for sequence end, and <UNK> for out-of-vocabulary tokens. All tokens appearing at least once in the training corpus were included. The English vocabulary contained 3,407 unique tokens and the Dzonglish vocabulary contained 3,311 unique tokens. The vocabularies are kept separate for each direction of translation.

3.5 Model Architectures

Five models were implemented and evaluated. The first four are trained from scratch on the collected corpus. The fifth uses pretrained multilingual weights fine-tuned on the corpus.

Seq2Seq GRU. The baseline model follows the encoder-decoder architecture of Cho et al. [8]. The encoder is a single-layer GRU that processes the source sequence token by token, producing a sequence of hidden states and a final context vector. The decoder is a single-layer GRU that generates the target sequence one token at a time using the encoder context. At each decoding step, the model predicts the next token from the decoder hidden state using a linear projection to the target vocabulary. At test time, the model uses greedy decoding.

Seq2Seq with Teacher Forcing. This model uses the same architecture as the vanilla Seq2Seq but applies teacher forcing during training. Rather than feeding the model's own predicted token as input to the next decoding step, the ground truth token is supplied with probability 0.5. Teacher forcing stabilizes training by preventing error accumulation during the early stages of learning, particularly beneficial when the model has not yet learned reliable token predictions [11].

Seq2Seq with Attention. This model augments the encoder-decoder framework with an attention mechanism following Bahdanau et al. [9]. At each decoding step, the attention module computes a weighted sum over all encoder hidden states, where the weights reflect the relevance of each source token to the current decoding position. The context vector produced by attention, together with the decoder hidden state and token embedding, is used to predict the output token. This allows the model to focus on the most relevant parts of the source sentence at each step, rather than relying on a single fixed-length context vector.

Transformer. The transformer model follows the architecture of Vaswani et al. [10]. Given the small dataset size, we used a reduced configuration with 2 encoder layers, 2 decoder layers, 4 attention heads, an embedding dimension of 128, and a feedforward dimension of 256. Positional encodings are added to the token embeddings to provide sequence order information. The model was trained with the Noam learning rate schedule [10], which linearly warms up the learning rate for 400 steps before decaying it proportionally to the inverse square root of the step number. This warmup schedule is essential for stable transformer training and was not applied to the recurrent models.

MarianMT (opus-mt-en-mul and opus-mt-mul-en). Two pretrained MarianMT models from the Helsinki-NLP group [15] were fine-tuned for the two translation directions. The opus-mt-en-mul model translates from English into multiple target languages and was fine-tuned for the English to Dzonglish direction. The opus-mt-mul-en model translates from multiple source languages into English and was fine-tuned for the Dzonglish to English direction. Each model contains approximately 143 million parameters. Fine-tuning was performed for 30 epochs using AdamW optimization with a learning rate of $5e-5$ and a OneCycleLR schedule. Beam search with 4 beams was used during inference. The MarianMT models benefit from pretraining on large multilingual corpora: while Dzonglish itself is not represented, the shared vocabulary space and multilingual representations learned during pretraining provide useful transfer for romanized and code-mixed language inputs.

All from-scratch models used an embedding dimension of 128, a hidden dimension of 256, one recurrent layer, and a dropout rate of 0.4. Label smoothing with a coefficient of 0.1 was applied to the cross-entropy loss for all from-scratch models, which improves calibration and reduces overconfident predictions on small datasets. Gradient clipping at 1.0 was applied during all training runs. Table 3 summarizes the hyperparameter settings.

Table 3. Hyperparameter settings for all models.

Hyperparameter	Seq2Seq	Seq2Seq+TF	Seq2Seq+Attn	Transformer	MarianMT
Embedding dim	128	128	128	128	512
Hidden dim / FF dim	256	256	256	256	2048
Layers (enc/dec)	1 / 1	1 / 1	1 / 1	2 / 2	6 / 6
Attention heads	-	-	Additive	4	8
Dropout	0.4	0.4	0.4	0.4	0.1
Teacher forcing ratio	0.0	0.5	0.5	-	-
Optimizer	Adam	Adam	Adam	Adam	AdamW
Learning rate	0.0005	0.0005	0.0005	Noam	$5e-5$
Epochs	50	50	50	50	30
Batch size	32	32	32	32	16
Label smoothing	0.1	0.1	0.1	0.1	-
Parameters (M)	~4.3	~4.3	~5.1	~3.8	~143

3.6 Evaluation Metrics

Six automatic evaluation metrics were used to assess translation quality. BLEU (Bilingual Evaluation Understudy) [19] measures the geometric mean of precision over n-gram overlaps (1-gram through 4-gram) between the hypothesis and reference, with a brevity penalty for short outputs. METEOR [20] evaluates translation quality using unigram recall and precision with

harmonic weighting and incorporates stemming and synonym matching, making it more robust to paraphrase than BLEU. chrF [21] is a character-level F-score metric that computes precision and recall over character n-grams (1 through 6). Character-level metrics are particularly informative for Dzonglish because the lack of standardized spelling means two valid translations of the same sentence may differ at the word level but share significant character-level overlap. ROUGE-1, ROUGE-2, and ROUGE-L [22] measure recall-oriented n-gram overlap between the hypothesis and reference. While primarily used in summarization evaluation, ROUGE provides complementary information about lexical coverage that is useful for characterizing low-resource translation systems. All metrics are reported as percentages (0-100 scale).

4. Results and Discussion

All experiments were conducted on Google Colaboratory using an NVIDIA Tesla T4 GPU with 16 GB GDDR6 VRAM. Training the four from-scratch models for 50 epochs across both translation directions took approximately 60 minutes, and fine-tuning each MarianMT model for 30 epochs took approximately 10 minutes per direction.

4.1 English to Dzonglish

Table 4 presents the evaluation results for English to Dzonglish translation. All four from-scratch models achieved BLEU scores of 0.00, indicating that the models were unable to generate translations with sufficient n-gram overlap at the 4-gram level to score above zero. This is consistent with the dataset size: standard NMT systems typically require tens of thousands of parallel sentences to achieve meaningful BLEU scores, whereas our training set contains fewer than 3,000 pairs. Despite the zero BLEU scores, the models show meaningful non-zero scores on character-level and unigram metrics. Seq2Seq with Attention achieved the highest scores among from-scratch models on METEOR (9.58), chrF (12.20), ROUGE-1 (18.82), and ROUGE-L (18.50), indicating that attention allows the model to produce translations with better lexical coverage than the baseline.

Table 4. Evaluation results for English to Dzonglish translation.

Model	BLEU	METEOR	chrF	ROUGE-1	ROUGE-2	ROUGE-L	Val Loss
Seq2Seq	0.00	7.77	11.17	16.46	2.20	16.16	5.893
Seq2Seq+Teacher Forcing	0.00	7.54	10.52	15.10	2.01	14.58	5.925
Seq2Seq+Attention	0.00	9.58	12.20	18.82	3.24	18.50	5.888
Transformer	0.00	3.95	4.90	3.08	0.40	3.06	6.074
MarianMT (opus-mt-en-mul)	5.24	21.29	29.82	32.05	9.28	30.27	3.079

The Transformer trained from scratch performed worst among all models in the English to Dzonglish direction, achieving METEOR of only 3.95 and chrF of 4.90. Examination of the sample outputs reveals a mode collapse behavior: the model repeatedly generates the most frequent Dzonglish token "nga" (meaning "I") regardless of the input. This occurs because the transformer requires substantially more data to learn reliable attention patterns from scratch. With fewer than 3,000 training pairs and a vocabulary of 3,311 tokens, the transformer lacks sufficient data density to learn meaningful cross-attention weights, causing it to default to high-frequency token generation. The Noam warmup scheduler prevented the instability observed in earlier training runs, but the data bottleneck remains the primary limiting factor.

MarianMT (opus-mt-en-mul) substantially outperformed all from-scratch models, achieving BLEU 5.24, METEOR 21.29, chrF 29.82, and ROUGE-1 32.05. The pretrained model benefits from multilingual representations learned across more than 1,000 language pairs during pretraining. Although Dzonglish is not among the pretrained languages, the romanized nature of Dzonglish means its tokens overlap significantly with Latin-script languages already present in the OPUS corpus. This cross-lingual transfer allows the model to generate more coherent translations after relatively brief fine-tuning. The sample output "tashi wong" for input "tashi will come" and "ani chim gaji mo" for "whose house is this?" illustrates that the model captures partial translation patterns successfully.

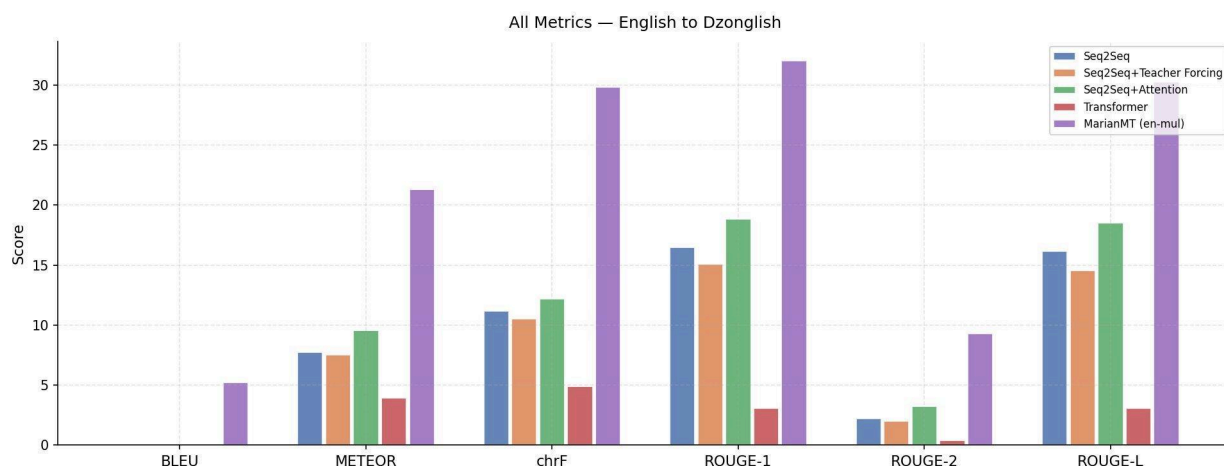


Figure 2. All metric scores for English to Dzonglish translation. MarianMT (en-mul) consistently outperforms all from-scratch models across every metric. Seq2Seq+Attention achieves the best performance among from-scratch models.

4.2 Dzonglish to English

Table 5 presents results for Dzonglish to English translation. The DZ-EN direction proves slightly easier for from-scratch models than EN-DZ, which is consistent with findings in related low-resource translation research where translation into a higher-resource language with a larger pretraining footprint benefits from more regularized target-side patterns. Seq2Seq with Attention is the only from-scratch model to achieve a non-zero BLEU score of 1.14, and also leads on

METEOR (12.63), ROUGE-1 (22.28), and ROUGE-L (21.80). The attention mechanism's ability to align specific Dzonglish tokens with their English counterparts explains this advantage.

Table 5. Evaluation results for Dzonglish to English translation.

Model	BLEU	METEOR	chrF	ROUGE-1	ROUGE-2	ROUGE-L	Val Loss
Seq2Seq	0.00	9.18	8.69	17.32	1.30	16.78	5.848
Seq2Seq+Teacher Forcing	0.00	10.19	8.82	17.52	1.69	16.87	5.875
Seq2Seq+Attention	1.14	12.63	10.12	22.28	3.47	21.80	5.851
Transformer	0.00	5.13	7.59	5.12	0.68	5.04	5.521
MarianMT (opus-mt-mul-en)	3.59	17.21	21.17	27.11	7.09	25.53	2.453

The Transformer again shows poor translation quality in the DZ-EN direction, producing repetitive phrase sequences such as "i am sure i am sure..." rather than meaningful translations. This mode collapse differs from the EN-DZ direction where "nga" was the dominant repeated token: in DZ-EN, the model appears to latch onto a high-frequency English phrase pattern. Both cases are symptomatic of insufficient training data for a parameter-rich architecture.

MarianMT (opus-mt-mul-en) achieves BLEU 3.59 and METEOR 17.21 for DZ-EN, which is lower than its performance in the EN-DZ direction. This asymmetry may reflect that Dzonglish as a source language is more challenging for the pretrained model to process than English, given the lack of explicit Dzonglish representation in the OPUS pretraining corpus. Nevertheless, the model demonstrates partial translation capability, generating "we are all set" for input "gey cc nga chey" (actual: "we won") and "what is it?" for "ga gi chim mo" (actual: "whose house is this?"). These outputs indicate the model understands the interrogative and declarative structure of inputs even when specific lexical mappings remain imprecise.

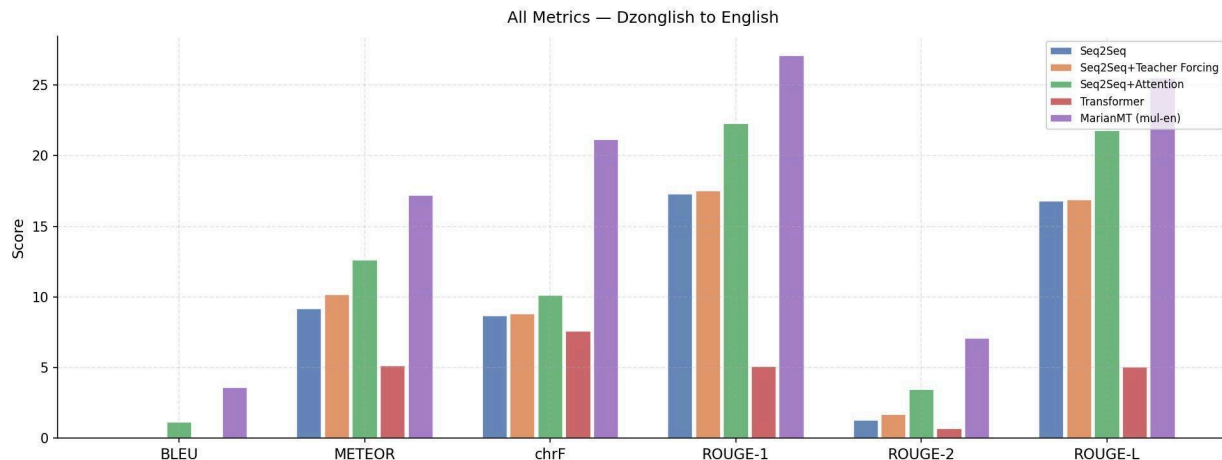


Figure 3. All metric scores for Dzonglish to English translation. MarianMT (mul-en) leads across all metrics. Among from-scratch models, Seq2Seq+Attention achieves the best results on METEOR, ROUGE-1, and ROUGE-L.

4.3 BLEU Score Comparison

Figure 4 provides a side-by-side BLEU comparison across all models and both directions. The chart clearly illustrates the performance gap between pretrained and from-scratch approaches on low-resource data. Among from-scratch models, only Seq2Seq with Attention achieves a non-zero BLEU score (1.14) and only in the DZ-EN direction. MarianMT achieves BLEU 5.24 for EN-DZ and BLEU 3.59 for DZ-EN. While these absolute BLEU scores are low by the standards of high-resource translation, they represent a substantial improvement over from-scratch baselines and are consistent with reported BLEU scores for other low-resource language pairs with similar corpus sizes in the literature.

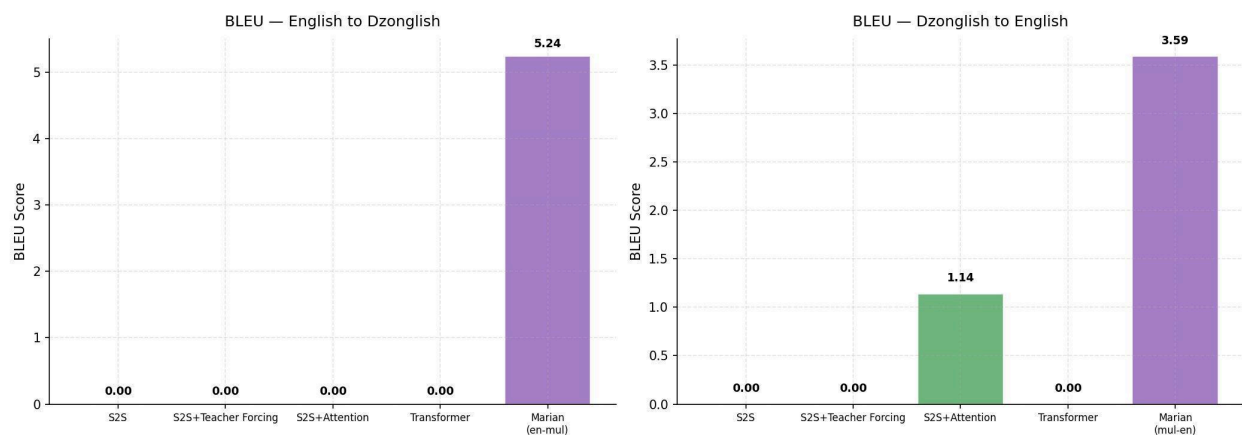


Figure 4. BLEU score comparison for all five models in both translation directions. MarianMT is the only approach to achieve meaningful BLEU scores, with EN-DZ (5.24) outperforming DZ-EN (3.59). All from-scratch models score 0.00 on BLEU except Seq2Seq+Attention on DZ-EN (1.14).

4.4 Training Dynamics

Figures 5 and 6 show the validation accuracy curves for from-scratch models in both translation directions. For English to Dzonglish, all three Seq2Seq variants converge to a validation accuracy of approximately 21%, stabilizing by epoch 10 with minimal improvement thereafter. The Transformer trails consistently at around 18% due to insufficient data to learn meaningful attention patterns. For Dzonglish to English, the Transformer actually achieves the highest validation accuracy of approximately 28% despite producing poor translations, suggesting that token-level accuracy is a misleading indicator of translation quality when the vocabulary distribution is heavily skewed toward frequent tokens.

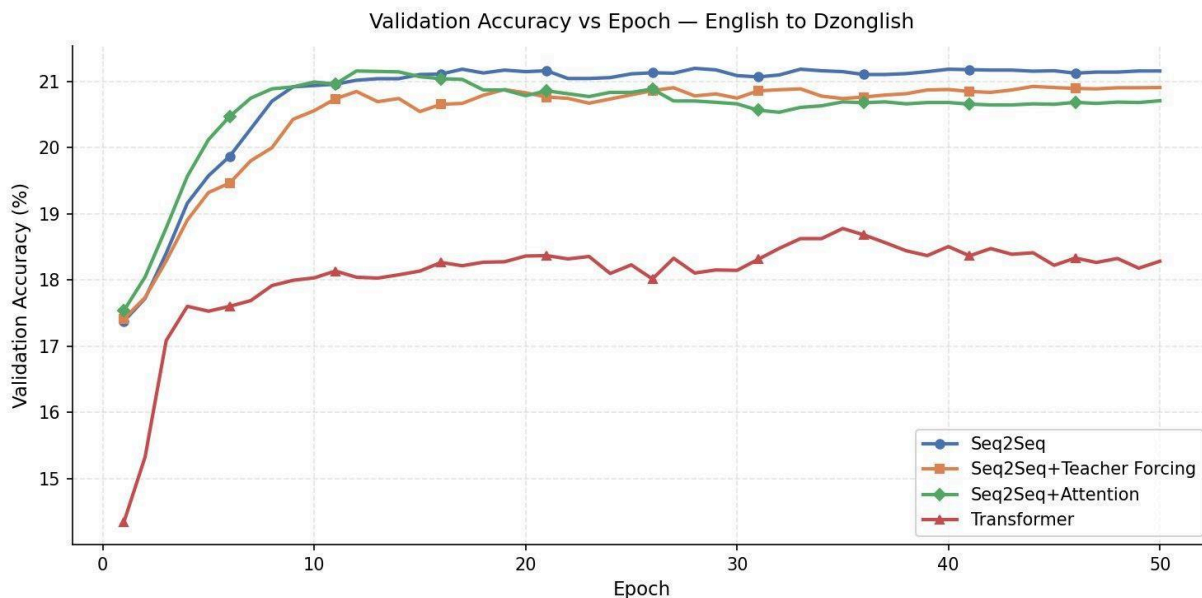


Figure 5. Validation accuracy vs epoch for from-scratch models, English to Dzonglish direction. All Seq2Seq variants plateau at around 21% by epoch 10. The Transformer consistently underperforms on this low-resource task.

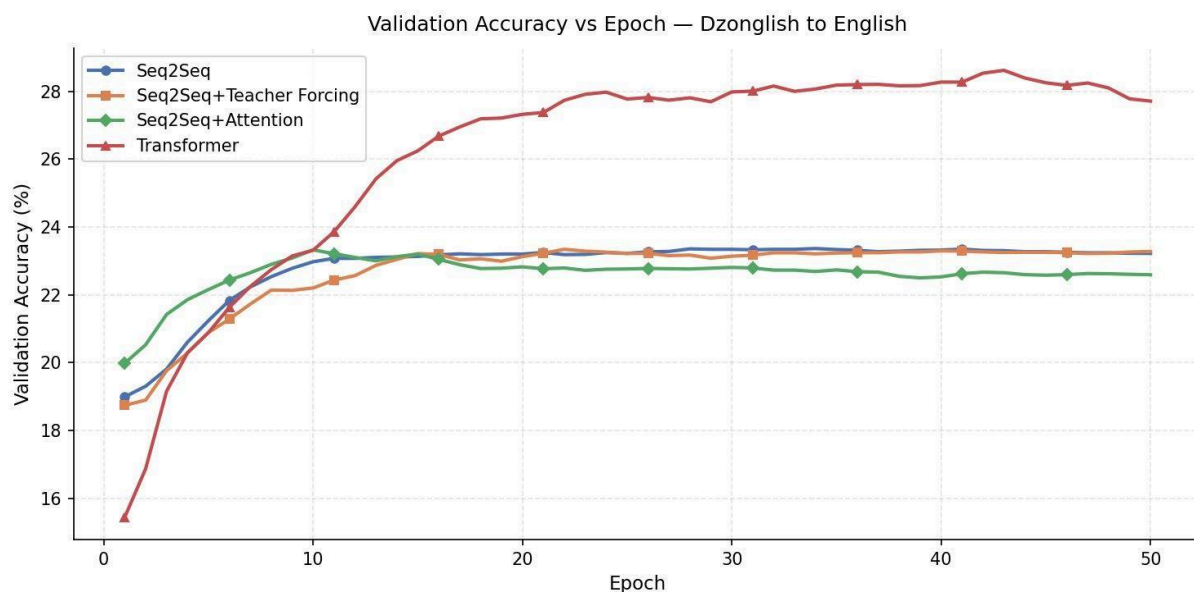


Figure 6. Validation accuracy vs epoch for from-scratch models, Dzonglish to English direction. The Transformer achieves the highest token-level accuracy (~28%) but produces the lowest translation quality metrics, indicating that high token accuracy can be misleading when models collapse to frequent token prediction.

Figures 7 and 8 show the training and validation loss curves for all five models. The MarianMT curves are visually distinct from the from-scratch models: training loss drops steeply from epoch 1 and reaches near-zero by epoch 15, indicating rapid adaptation of pretrained weights to the new domain. Validation loss for MarianMT reaches a minimum around epoch 5 and then rises gradually, a sign of overfitting given the small dataset. Early stopping via saving

the best checkpoint prevents this overfitting from affecting final evaluation. The from-scratch models show higher training and validation losses throughout, with validation loss increasing slightly as training progresses, consistent with mild overfitting on the small training set.

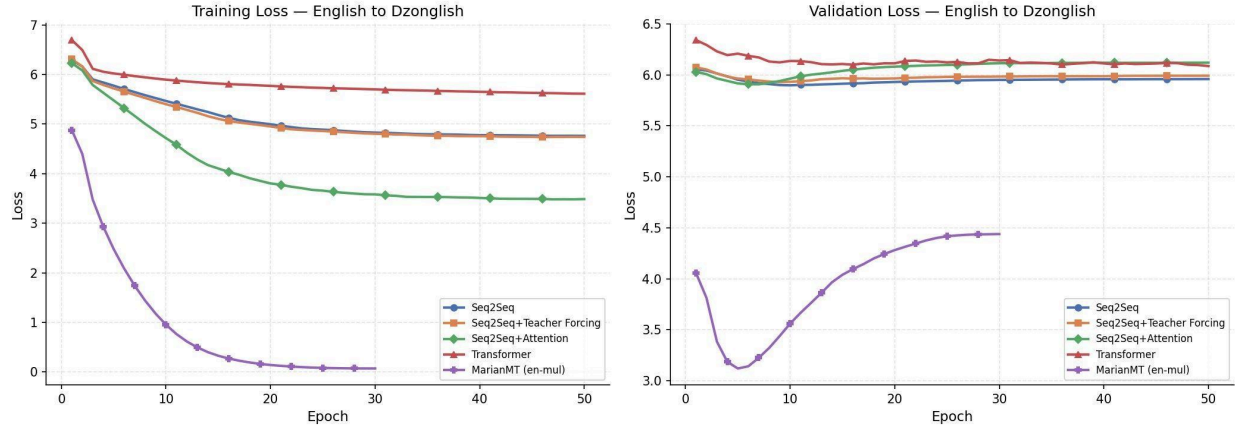


Figure 7. Training loss (left) and validation loss (right) for all models, English to Dzonglish direction. MarianMT training loss drops sharply to near-zero, reflecting rapid fine-tuning of pretrained weights. From-scratch models show slower convergence and higher final losses.

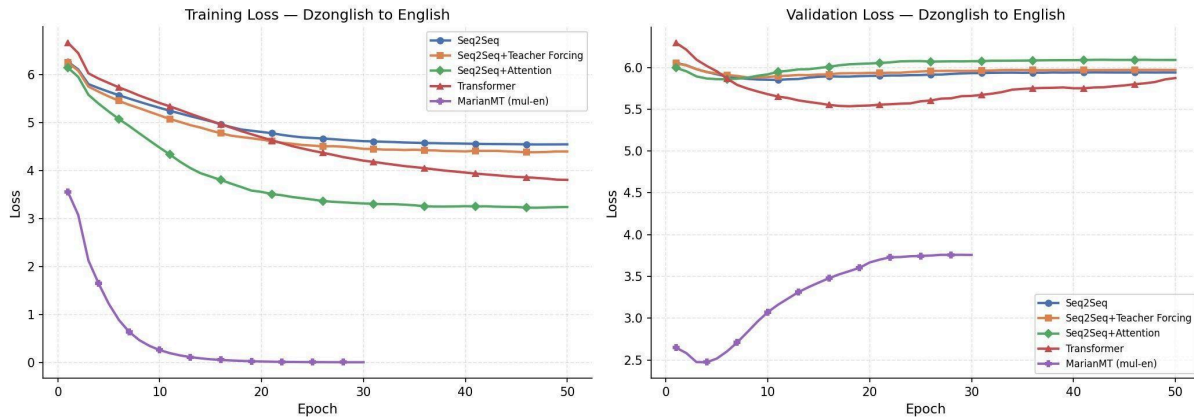


Figure 8. Training loss (left) and validation loss (right) for all models, Dzonglish to English direction. The pattern mirrors the EN-DZ direction: MarianMT fine-tuning converges rapidly while from-scratch models plateau at significantly higher loss values.

4.5 Sample Translation Analysis

Table 6 presents qualitative translation examples from the test set across all five models for both directions. The examples reinforce the quantitative findings. From-scratch Seq2Seq models produce partial lexical matches but fail to generate grammatically coherent or semantically complete outputs. Seq2Seq with Attention performs best among this group, producing "nga chey bay" for "we won" which shares the token "chey" with the reference "gey cc nga chey". The Transformer collapses to repetitive token generation in both directions. MarianMT produces the most coherent outputs: "tashi wong" for "tashi will come" demonstrates

correct proper noun preservation and verb translation, while "we are all set" for the DZ-EN direction captures the declarative sentiment of the source even if the specific lexical mapping is imprecise.

Table 6. Sample translations from selected models for both directions.

Direction	Input	Reference	Seq2Seq+Attn	MarianMT
EN-DZ	whose house is this?	ga gi chim mo	nga gi dhi dhi	ani chim gagi mo
EN-DZ	tashi will come.	tashi nga wong	tashi gi na	tashi wong
EN-DZ	we won.	gey cc nga chey	nga chey bay	we gay
DZ-EN	ga gi chim mo	whose house is this?	i is the her	what is it?
DZ-EN	tashi nga wong	tashi will come.	i am not to	i am tashi.
DZ-EN	gey cc nga chey	we won.	i am not we	we are all set.

4.6 Discussion

The results collectively demonstrate two key findings. First, the low-resource nature of Dzonglish translation severely limits from-scratch NMT models. With fewer than 3,000 training pairs and vocabularies of over 3,300 tokens, models face extreme data sparsity. The attention mechanism provides the most consistent benefit among from-scratch approaches, confirming that selective source-token focusing is more valuable than teacher forcing or increased model capacity for this data size. Second, pretrained multilingual models provide substantial gains even on a language pair not directly represented in pretraining. The 143M-parameter MarianMT models leverage cross-lingual transfer from romanized and multilingual representations, outperforming all from-scratch models by wide margins on every metric.

The asymmetry between EN-DZ and DZ-EN results is also noteworthy. MarianMT performs better in the EN-DZ direction (BLEU 5.24 vs 3.59), while Seq2Seq with Attention performs better in DZ-EN (BLEU 1.14 vs 0.00). This suggests that English is a more effective source language for both pretrained and from-scratch models, likely because English is better represented in both the pretraining corpus and the model vocabulary.

The chrF metric consistently provides the most nuanced differentiation between models among the from-scratch systems. Because chrF operates at the character level, it captures partial spelling matches that are common when a model produces a related but not identical Dzonglish word. This makes chrF particularly well-suited for evaluating Dzonglish translation, where orthographic variation is a fundamental property of the language rather than a translation error.

5. Conclusion

This paper presents the first study on neural machine translation for bidirectional translation between English and Romanized Dzongkha (Dzonglish). We constructed a parallel corpus of 3,813 sentence pairs through crowdsourced data collection, applied a targeted preprocessing pipeline suited to the linguistic properties of Dzonglish, and evaluated five neural machine translation systems: Seq2Seq GRU, Seq2Seq with Teacher Forcing, Seq2Seq with Attention, Transformer, and fine-tuned MarianMT models in both translation directions.

Our experiments demonstrate that from-scratch models struggle on this low-resource dataset, with Seq2Seq with Attention emerging as the best performer among them. Pretrained MarianMT models achieve substantially better results through cross-lingual transfer, attaining BLEU 5.24 and METEOR 21.29 for English to Dzonglish and BLEU 3.59 and METEOR 17.21 for Dzonglish to English. The character-level chrF metric proves particularly informative for evaluating Dzonglish translation given the orthographic variability inherent in the language.

Several directions remain for future work. Expanding the parallel corpus to 15,000 or more pairs through continued crowdsourcing would directly address the primary performance bottleneck. Developing a standardized Dzonglish romanization scheme through collaboration with linguists and language authorities could reduce orthographic noise across the dataset. Investigating larger pretrained models such as mT5 and mBART for fine-tuning on this task may yield further improvements. Subword tokenization methods such as SentencePiece or Byte Pair Encoding could reduce out-of-vocabulary rates and better handle the morphological characteristics of Dzonglish. Finally, human evaluation by native Dzonglish speakers would provide richer quality assessment than automatic metrics alone, particularly for a language where spelling standardization is still emerging.

The dataset, model code, evaluation scripts, and pretrained model checkpoints are publicly available at <https://github.com/aploslab/Dotu-dzonglish-nmt.git> to support future research on Dzonglish NLP and low-resource language technology for Bhutanese languages.

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